

التغذية العلاجية
والسمنة



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من فضلك اضغط هنا..

Management of Regional Obesity

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Regional obesity

- **What is regional obesity?**

The traditional definition of obesity includes a measure of height/weight and defined as Body Mass Index (BMI). Now with the advancement of science we are realizing regional obesity is much more important for health. Here we measure the girth of waist and hip and express that as ratio called **WAIST-HIP RATIO (WHR)**. For men normal is less than 0.85 and for women it is less than 0.90.

Definitions

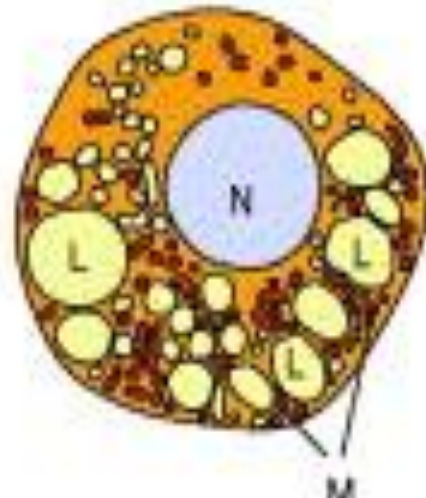
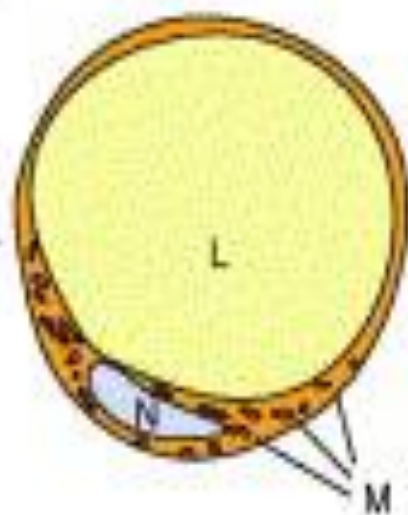
- 1. An **excess of subcutaneous fat in proportion to lean body mass**. Excess fat accumulation is associated with increase in the size (hypertrophy) as well as the number (hyperplasia) of adipose tissue cells. These do not differentiate between excess adiposity and increased lean body mass. In contrast, subscapular and triceps skinfold measurements and determination of the waist:hip ratio help define the regional deposition of fat and differentiate the more medically significant central obesity from peripheral obesity in adults.
- **No single cause can explain all cases of obesity**. Ultimately it results from an imbalance between energy intake and energy expenditure. Although faulty eating habits related to failure of normal satiety feedback mechanisms may be responsible for some cases, many obese people neither consume more calories nor eat different proportions of foodstuffs than nonobese persons. Contrary to popular belief, obesity is not caused by disorders of pituitary, thyroid, or adrenal gland metabolism. However, it is often associated **with hyperinsulinism** and relative insulin resistance.
- Studies of obese twins strongly suggest the presence of **genetic influences** on resting metabolic rate, feeding behavior, changes in energy expenditures in response to overfeeding, lipoprotein lipase activity, and basal rate of lipolysis.
- Environmental factors associated with obesity include socioeconomic status, **race**, region of residence, season, urban living, and being part of a smaller family.

What is the relation with general obesity?

- In most of the cases it does not go hand in hand. You may have a normal BMI but your WHR may not be. So normal height-weight ratio or maintaining an ideal body weight is not always reassuring.
- How does it happen? The fat in the abdomen are resistant to the action of Insulin, more so in comparison to the other part of the body. With an excess of abdominal fat Insulin resistance develops which increases Insulin resistance further and the cycle continues. This **Insulin resistance** is the main reason of adverse health outcome with regional obesity.

What are the health consequences of regional obesity?

- Regional obesity is associated with a cluster of metabolic problems such as Diabetes, high blood pressure, high cholesterol, atherosclerosis which all lead to problems with the heart (heart attack) and brain (stroke). Medically we called this Syndrome X. The other concern is back pain, because of the pot-belly the forward curvature of the lumbar spine increases and is one of the risk of disc propblem and wear and tear leading to the spondylitis.

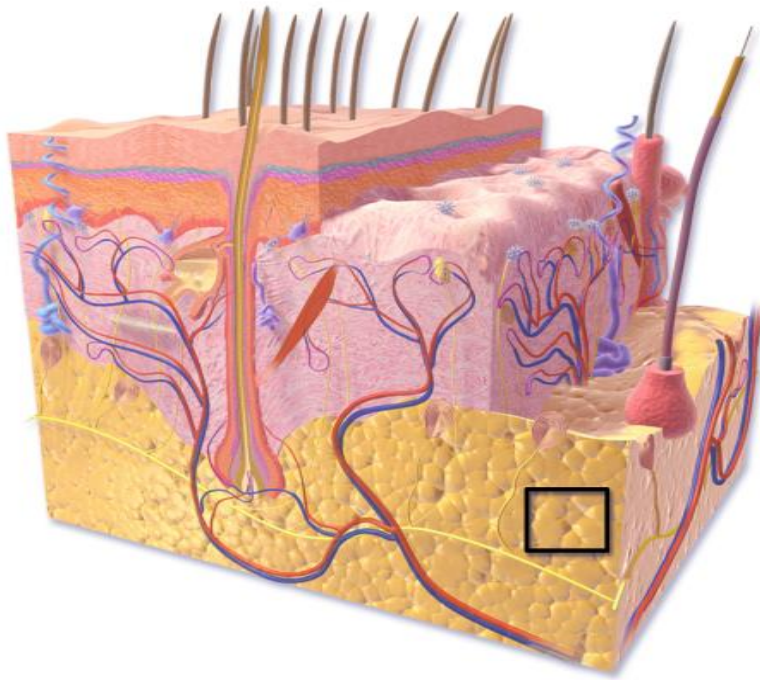


White adipocyte

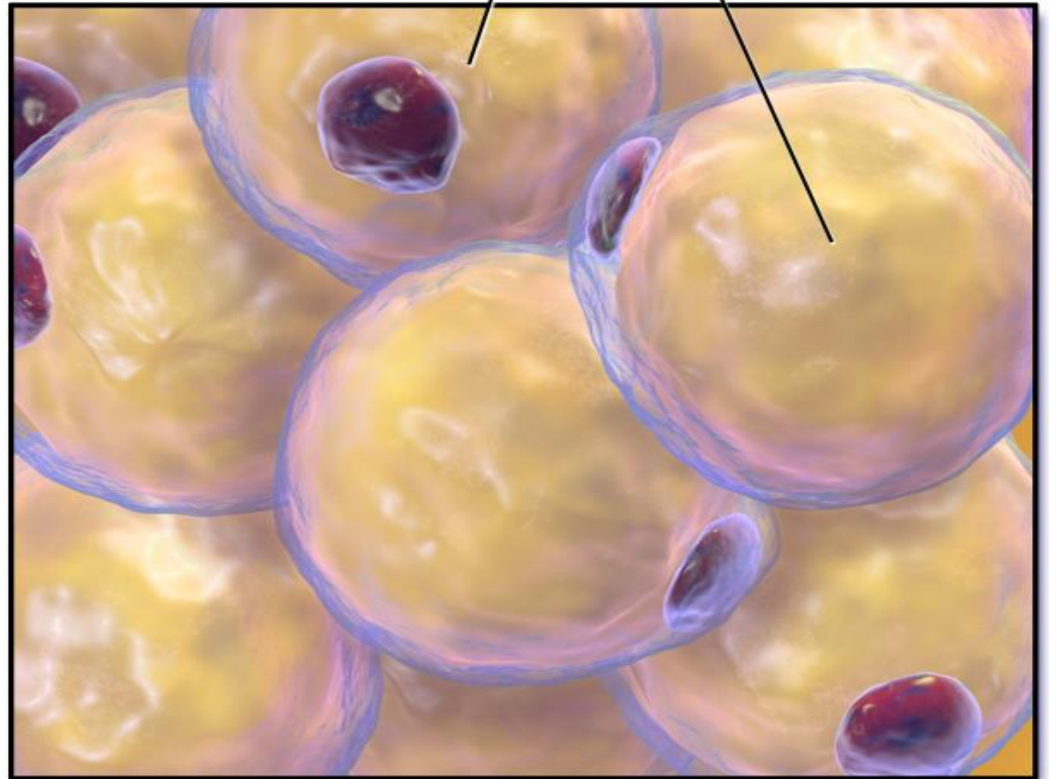
- Lipids (L) in a single vacuola
- Peripheral nucleus (N)
- Few mitochondria (M), poorly developed
- No *UCP1* gene expression

Brown adipocyte

- Lipids (L) in multiple vacuolae
- Central nucleus (N)
- Many mitochondria (M), highly developed
- *UCP1* gene expression



Adipocytes
(*white adipose cells*)



Adipose Tissue

What is the treatment?

- First and most important step is to make us aware that **abdominal obesity** is bad not only for cosmetic reason but also for health. **Change of life style** with so-called “modernisation” of our society (less physical work, more and more fast food intake with high fat and refined carbohydrate) has the burden of blame to share. We are using the term **“Therapeutic Life Style (TLC)” changing** as the most important measure. We need to eat the correct food, which obviously means to avoid high fat and calorie fast food as much as possible. More importantly realizing what is healthy eating, like eating plenty of vegetables and optimum fruits on daily basis. We need to undertake some form of physical exercise every day for at least 30-45 min. Also we need to keep an eye on our BMI, WHR, blood pressure, cholesterol, etc on a regular basis.

Modalities of treatment:

- Mesotherapy.
- Cavitation ultrasound.
- RF.
- Carboxy therapy.
- Low level laser.
- Cryolipolysis.

cryolipolysis

- Cryolipolysis is the non-invasive cooling of adipose tissue to induce lipolysis - the breaking down of fat cells - to reduce body fat without damage to other tissues.



The scientific principles of Cryolipolysis

- Discovered by Dermatologists **Dieter Manstein, MD, PhD and R. Rox Anderson, MD, of the Wellman Center** for Photomedicine at General Hospital in Boston, Harvard Medical School. The physicians and their team conducted research that demonstrated, subcutaneous fat cells are naturally more vulnerable to the effects of cold than other surrounding tissue. Their initial work, published in Lasers in Surgery and Medicine in **November 2008**.

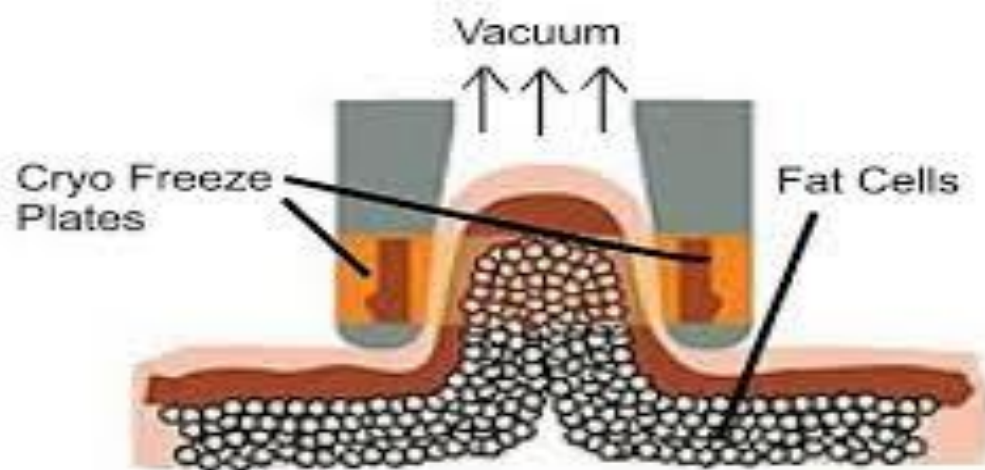


Mechanism of Action

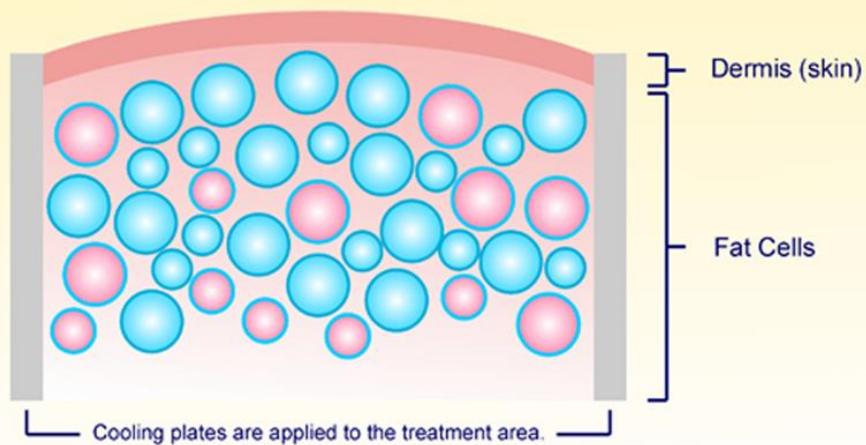
- Exposure to cooling via energy extraction causes fat **cell apoptosis** - a natural, controlled cell death, which leads to the release of cytokines and other inflammatory mediators that gradually eliminate the affected cells.
- **Inflammatory cells** gradually digest the affected fat cells in the months after the procedure, reducing the thickness of the fat layer.
- **Lipids** from the fat cells are slowly released and transported by the lymphatic system to be processed and eliminated, much like that of fat from food.



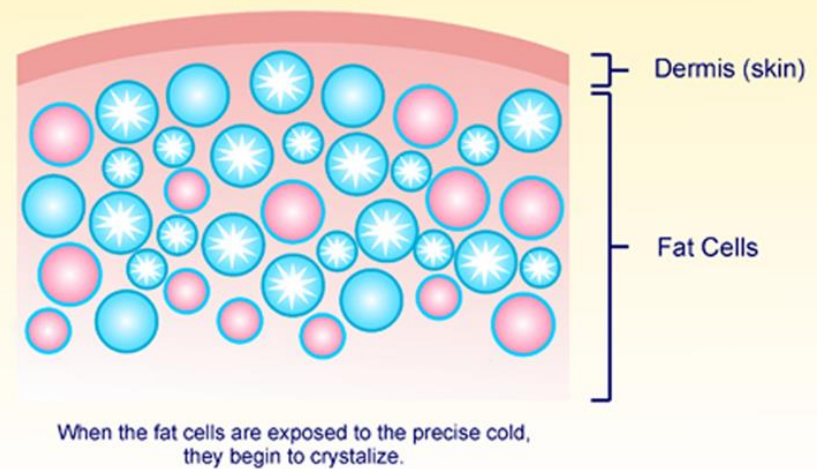
Cryo Lipolysis



Day of procedure



Immediately following the procedure





- **Side effects**

Side effect data are based on a limited experience. Transient **local redness, bruising and numbness of the skin** are common side effects of the treatment and are expected to subside. **Typically sensory deficits will subside within a month.** The effect on peripheral nerves was investigated and failed to show permanent detrimental results. **No serious long-lasting side effects were encountered during follow-up time of six months.** Scarring, ulcerations, or changes in **blood lipids or liver functions have not been reported.**

- **FDA approval**

In September 2010, the [Food and Drug Administration](#) approved the Cryolipolysis device. [Zeltiq Aesthetics](#) of Pleasanton, California. It had already been approved to cool skin for pain reduction in selected applications and was used [off-label](#) prior to FDA approval for cryolipolysis.

Enhanced clinical outcome with manual massage following cryolipolysis treatment: a 4-month study of safety and efficacy.

[Boey GE¹](#), [Wasilenchuk JL](#).

Abstract

BACKGROUND AND OBJECTIVES:

Cryolipolysis procedures have been shown to safely and effectively reduce the thickness of fat in a treated region. This study was conducted to determine whether the addition of post-treatment manual massage would improve efficacy while maintaining the safety profile of the original cryolipolysis treatment protocol.

MATERIALS AND METHODS:

The study population consisted of an efficacy group (n = 10) and a safety group (n = 7). Study subjects were treated on each side of the lower abdomen with a Cooling Intensity Factor of 42 (-72.9 mW/cm²) for 60 minutes. One side of the abdomen was massaged post-treatment and the other side served as the control. Immediately post-treatment, the massage side was treated for 1 minute using a vigorous kneading motion followed by 1 minute of circular massage using the pads of the fingers. For the efficacy group, photos and ultrasound measurements were taken at baseline, 2 months, and 4 months post-treatment. For the safety group, histological analysis was completed at 0, 3, 8, 14, 30, 60, and 120 days post-treatment to examine the effects of massage on subcutaneous tissue over time.

RESULTS:

Post-treatment manual massage resulted in a consistent and discernible increase in efficacy over the non-massaged side. At 2 months post-treatment, mean fat layer reduction was 68% greater in the massage side than in the non-massage side as measured by ultrasound. By 4 months, mean fat layer reduction was 44% greater in the massage side. Histological results showed no evidence of necrosis or fibrosis resulting from the massage.

CONCLUSION:

Post-treatment manual massage is a safe and effective technique to enhance the clinical outcome from a cryolipolysis procedure.

- [Lasers Surg Med.](#) 2014 Feb;46(2):75-80. doi: 10.1002/lsm.22207. Epub 2013 Dec 3.

Three-dimensional volumetric quantification of fat loss following cryolipolysis.

- [Garibyan L¹](#), [Sipprell WH 3rd](#), [Jalilian HR](#), [Sakamoto FH](#), [Avram M](#), [Anderson RR](#).
- [Author information](#)

Abstract

BACKGROUND AND OBJECTIVES:

Cryolipolysis is a noninvasive and well-tolerated treatment for reduction of localized subcutaneous fat. Although several studies demonstrate the safety and efficacy of this procedure, volumetric fat reduction from this treatment has not been quantified. This prospective study investigated the change in volume of fat after cryolipolysis treatment using three-dimensional (3D) photography.

MATERIALS AND METHODS:

A prospective study of subjects treated with cryolipolysis on the flank (love handle) was performed at Massachusetts General Hospital. Volume measurements were performed with a Canfield Scientific Vectra three-dimensional camera and software to evaluate the amount of post procedure volume change. Clinical outcomes were assessed with caliper measurements, subject surveys, and blinded physician assessment of photographs.

RESULTS:

Eleven subjects were enrolled in this study. Each subject underwent a single cycle of cryolipolysis to one flank. The untreated flank served as an internal control. The follow-up time after treatment was 2 months. The mean amount of calculated absolute fat volume loss using 3D photography from baseline to 2 months follow-up visit was 56.2 ± 25.6 from the treatment site and 16.6 ± 17.6 cc from the control ($P < 0.0001$). A mean absolute difference of 39.6 cc between the treated and untreated sides was calculated at 2 months post-treatment. Comparison of caliper measurements from baseline to 2 months post-treatment demonstrated significant reduction of the treated flank from 45.6 ± 5.8 mm at baseline to 38.6 ± 4.6 mm at 2 months post-treatment ($P < 0.001$). The untreated flank did not show significant reduction with caliper measurements demonstrating 45.3 ± 5.0 mm at baseline and 44.6 ± 5.1 mm at 2 months post-treatment ($P = 0.360$). No unexpected side effects or adverse events were reported. Post-treatment satisfaction surveys demonstrated 82% of subjects were satisfied with the results.

CONCLUSIONS:

Cryolipolysis is a safe, well-tolerated, and effective noninvasive fat removal methodology that on average leads to 39.6 cc of fat loss of the treated flank at 2 months after a single treatment cycle.

Noninvasive selective cryolipolysis and reperfusion recovery for localized natural fat reduction and contouring.

- [Sasaki GH¹](#), [Abelev N](#), [Tevez-Ortiz A](#).
- [Author information](#)

Abstract

BACKGROUND:

Cryolipolysis is a contemporary method of reducing fat by controlled extraction of heat from adipocytes.

OBJECTIVES:

The authors recorded temperature profiles during a single cryolipolysis treatment/recovery cycle (with and without massage) and report on the clinical safety and efficacy of this procedure

METHODS:

In the pilot study group (PSG), the abdomens of 6 patients were treated with cryolipolysis and subdermal temperatures were recorded. In the clinical treatment group (CTG), 112 patients were treated without temperature recordings and results were evaluated through matched comparison of standardized photographs, caliper measurements, ultrasound imaging, and global assessments.

RESULTS:

Thirty minutes into the cooling phase, subdermal temperatures of patients in the PSG declined precipitously from pretreatment levels and remained low until the end of treatment. During recovery, subdermal temperatures of the only subject who received massage returned faster and to higher levels than the temperatures of subjects who did not receive massage. Patients in the CTG who were available for follow-up measurements at 6 months (n = 85) demonstrated an average fat reduction of 21.5% by caliper measurements; 6 random patients from this group also showed an average of 19.6% fat reduction by ultrasound imaging at 6 months. Global assessments were highest for the abdomen, hip, and brassiere rolls. Minimal side effects were observed, and patients experienced no significant downtime.

CONCLUSIONS:

Noninvasive cryolipolysis results in a predictable and noticeable fat reduction within 6 months and does not cause skin damage. Profiling of subdermal temperatures may provide additional insights for improving clinical effectiveness and safety. Level of Evidence: 3.

Carboxytherapy mode of action

- Carbon Dioxide is a natural gas that exists in the body of every person. It is a byproduct of metabolism and it increases with the increase in exerted muscular effort. This means that when you practice physical exercises, the amount of carbon dioxide in your tissue will increase.
- The role of carbon dioxide is to dilate the blood vessels of the body in order to receive more oxygenated blood carrying nutrients to the muscles. This explains how the muscles grow after exercise. Carbon Dioxide also has another important role which is destroying the fat cells which are found in the subcutaneous layer of the skin.





THANK YOU